

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory**Lab 5 – C++ Programming: Function Overloading****Date:** 25.08.2026**Semester:** B.Tech 3rd Semester**Section:** CSE B2**Instructions**

- Write all programs in C++.
- Use **function overloading** to solve each problem.
- Use the same function name for the overloaded versions.
- Differentiate overloaded functions using the **number, type, or order of parameters**.
- Arrays and pointers may be used as function parameters wherever required.
- Display the output in a neat and readable format.
- Do not create separate function names for operations that are required to demonstrate overloading.

1. Number Calculator

Write a C++ program to perform a calculation using overloaded functions for:

- two integer values,
- three integer values,
- two floating-point values.

Display the result for each case.

***Hint:** Use the same function name and differentiate the functions through their parameter lists.*

2. Value Comparison

Write a C++ program to find the larger value using an overloaded function. The program should be able to compare:

- two integers,
- two floating-point numbers,
- three integers.

Display the appropriate result for each case.

***Hint:** The overloaded functions may differ in the number or type of parameters.*

3. Array Total

Write a C++ program to calculate the total of elements using an overloaded function. The program should work with:

- an integer array,
- a floating-point array,
- a portion of an integer array specified by the number of elements to consider.

Display the calculated total in each case.

Hint: Use the array type and/or additional parameters to distinguish the overloaded functions.

4. Element Search

Write a C++ program to search for an element using an overloaded function. The program should support:

- searching for an integer in an integer array,
- searching for a character in a character array,
- searching for an integer only within a specified range of an integer array.

Display the position of the element if it is found; otherwise, display an appropriate message.

Hint: Keep the function name the same while changing the parameter list.

5. Modify a Value

Write a C++ program to modify data using overloaded functions. The program should:

- add a specified value to an integer,
- add a specified value to a floating-point number,
- modify an integer value using its pointer.

Display the value before and after modification.

Hint: Pay attention to the difference between a normal variable and a pointer parameter.

6. Display Data

Write a C++ program using function overloading to display:

- an integer,
- a floating-point number,
- a character,
- all elements of an integer array,
- all elements of a character array.

Use a common function name for displaying all types of data.

Hint: The compiler should distinguish the functions using their parameter types or parameter lists.

7. Compare Data Sets

Write a C++ program using overloaded functions to compare:

- two integers,
- two floating-point numbers,
- two integer arrays of equal size.

For individual values, display the larger value. For arrays, determine whether both arrays contain identical elements.

Hint: The array version will require the array and its size as parameters.

8. Counting Operation

Write a C++ program using overloaded functions to perform the following:

- count the number of digits in an integer,
- count the number of elements in an integer array,
- count the occurrences of a given character in a character array.

Display the result of each operation.

Hint: Use the same function name but design different parameter lists for each operation.

9. Maximum Value Finder

Write a C++ program using function overloading to find the maximum value in the following cases:

- between two integers,
- between two values accessed through integer pointers,
- among all elements of an integer array using a pointer and its size.

Display the maximum value for each case.

Hint: Carefully distinguish between an integer parameter, an integer pointer, and a pointer with an additional size parameter.

10. Overloaded Data Processor

Write a C++ program that performs a common meaningful operation using overloaded functions for the following inputs:

- two integers,
- an integer and a floating-point value,
- two floating-point values,
- an integer array and its size,
- two integer pointers.

The operation should produce a meaningful result for every case. Demonstrate all overloaded versions from `main()`.

*Hint: Design the overloaded functions yourself. Carefully consider the **number, type, and order of parameters** when creating each version.*